

Principle 1 (Physics is the homotopy type). The modular homotopy type of a chiral algebra $\mathcal{A}$ —the formal moduli problem $\mathcal{M}_{\mathcal{A}}^{\text{mod}}(R) = \text{MC}(\text{Def}_{\infty}^{\text{mod}}(\mathcal{A}) \otimes R)/\text{gauge}$ —determines, through its finite-order projections, every perturbative observable of the corresponding holomorphic-topological QFT. The A_{∞} operation m_k is the k -point scattering amplitude; the genus tower $F_g = \kappa \cdot \lambda_g$ is the loop expansion; the R -matrix $r(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_{\mathcal{A}})$ is the scattering matrix; the Yangian is the symmetry algebra of the S -matrix; Koszul duality is S -duality; the shadow depth classification $\mathbf{G}/\mathbf{L}/\mathbf{C}/\mathbf{M}$ is the interaction complexity hierarchy (free/gauge/contact/gravitational).

The following dictionary records the systematic translation between perturbative QFT and modular Koszul theory. Each entry in the left column is a physical concept; the middle column gives its algebraic incarnation; the right column specifies the source within the MC framework.

Physics	Algebra	MC source
k -point scattering amplitude	A_{∞} product m_k	arity- k bar differential
genus- g vacuum amplitude	free energy $F_g(\mathcal{A}) = \kappa \cdot \lambda_g^{\text{FP}}$	scalar trace of $\Theta^{(g)}$
loop expansion	genus tower $\{F_g\}_{g \geq 1}$	$\hat{A}$ -genus generating function
S -matrix	collision R -matrix $r(z)$	$\text{Res}_{0,2}^{\text{coll}}(\Theta_{\mathcal{A}})$
S -matrix symmetry	Yangian $Y_h(\mathfrak{g})$	RTT presentation from $r(z)$
S -duality	Koszul duality $\mathcal{A} \leftrightarrow \mathcal{A}^!$	Verdier intertwining (Thm A)
line operators	$\mathcal{A}^!$ -mod	DK bridge (MC3)
anomaly cancellation	$\kappa_{\text{eff}} = \kappa(\text{matter}) + \kappa(\text{ghost}) = 0$	critical dimension
one-loop exactness	PBW / chirally Koszul	E_2 -collapse (K1–K2)
BV master equation	$D_{\mathcal{A}}^2 = 0$	bar-intrinsic MC (Thm B)
free / gauge / contact / gravitational	$\mathbf{G} / \mathbf{L} / \mathbf{C} / \mathbf{M}$	shadow depth r_{max}
perturbative finiteness	HS-sewing condition	analytic bar coalgebra (MC5)
complementarity / CPT	$Q_g(\mathcal{A}) + Q_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, Z(\mathcal{A}))$	Hodge decomposition (Thm C)
Coulomb moduli	$\text{MC}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}})$	MC moduli of $\Theta_{\mathcal{A}}$
gravitational Ward identity	shadow connection $\nabla_{g,n}^{\text{hol}}$	$d - \text{Sh}_{g,n}(\Theta_{\mathcal{A}})$